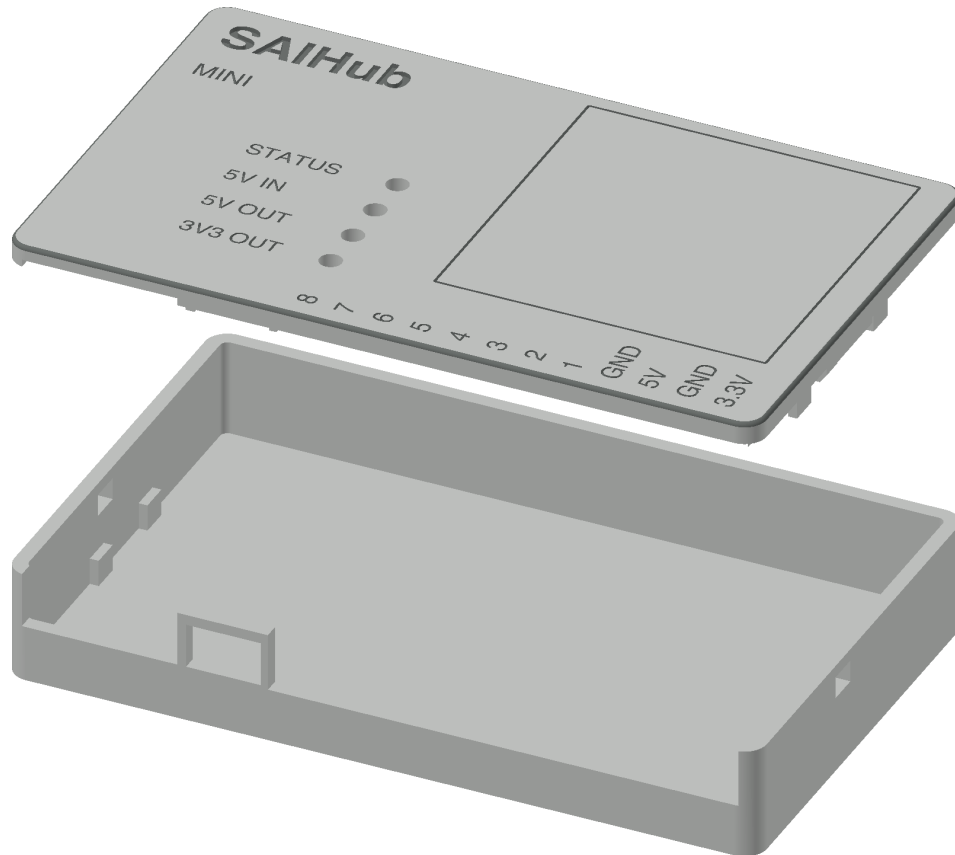


01 WITHOUT PCB

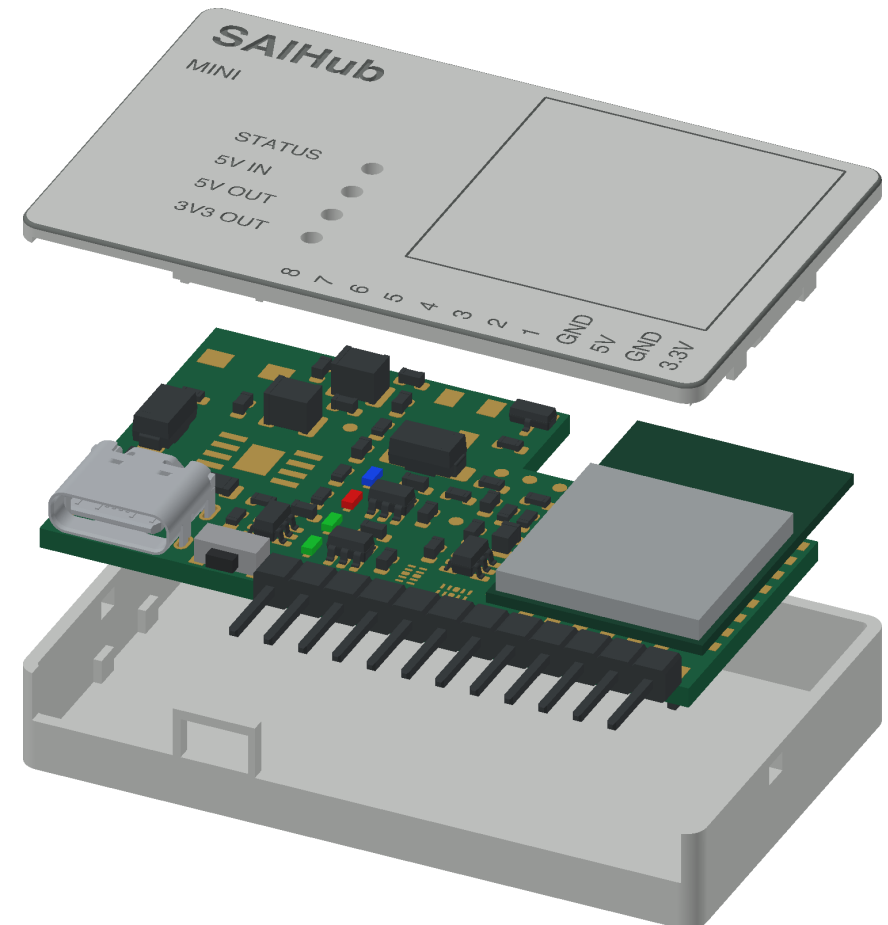


50.4 x 30.4 x 9 mm overall

9.0 mm body; QR outline and lettering recessed 0.3 mm

Four labelled top indicators; USB and button access through the side.

02 WITH PCB



Board: 48.0 x 28.0 x 1.6 mm, shown in assembly orientation.

Vertical separation is for illustration; XY alignment is retained.

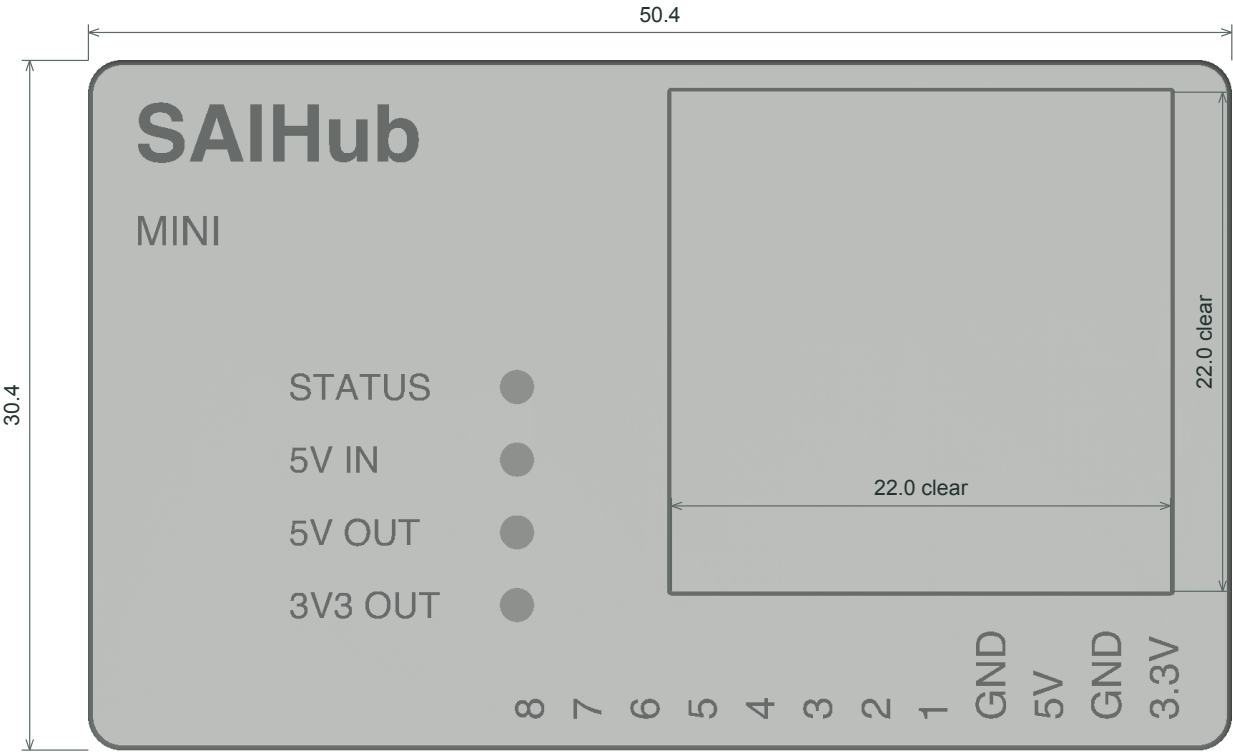
Available models shown; button, module and LEDs are illustrative.

Shell mass excludes the PCB, components and QR sticker.

SAIHub Mini

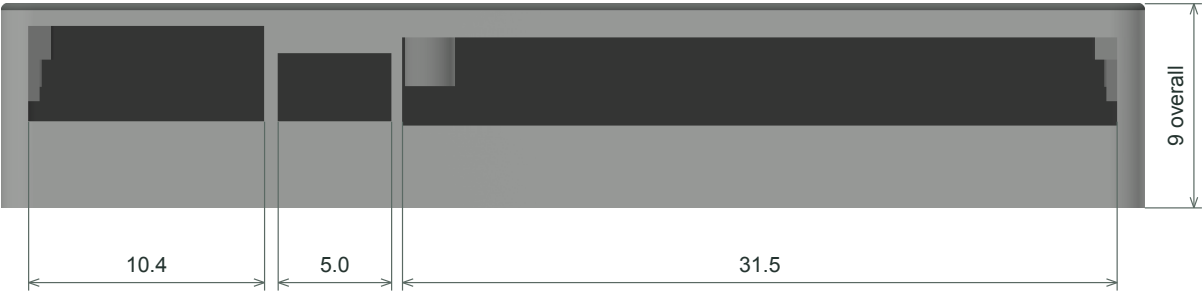
Three-view drawing

Third-angle arrangement | Scale 3:1 at 100% print size | Shell only



TOP

FRONT / CONNECTOR EDGE



DETAIL DIMENSIONS

- QR outline outside: 22.36 x 22.36
- Sticker bed: 22.0 x 22.0, flat
- Outline width: 0.18; recess: 0.3
- Text recess: 0.3 (roof remaining: 0.7)
- Nominal wall / floor / roof: 1.0
- Outer corner radius: R1.5
- Top edge round: R0.3
- Nesting overlap: 1.0
- Nesting clearance: 0.2 per side
- PCB clearance: 0.2 per side
- 2 side snaps; 0.25 engagement
- 4 indicator holes: diameter 1.5; pitch 3.2

Side opening heights:
USB 4.2 / button 3.0 / header 3.9

RIGHT SIDE



Dimensions govern. Views show external geometry; internal edges are omitted.

4.33 g

Nominal finished shell mass

Base + cover, ABS at 1.04 g/cm3

CALCULATION

Mass = CAD material volume / 1000 x ABS density

Total material volume: 4.15914 cm3

1 mm skins and walls assumed fully filled by extrusion.

The empty enclosure cavity is already excluded from volume.

PART	MATERIAL VOLUME	ESTIMATED ABS MASS
Base	2.369 cm3	2.46 g
Cover	1.790 cm3	1.86 g
TOTAL	4.159 cm3	4.33 g

PRINT ESTIMATE

Excludes supports, brim, purge, PCB, components and sticker.
Actual mass varies with filament density, line width, gaps and flow.
Do not multiply this thin-wall estimate by a general infill percentage.
Use the slicer's filament mass for the final chosen print settings.

Density source: PolyLite ABS Technical Data Sheet, V5.5, 1.04 g/cm3 at 23 C.

[Open manufacturer data sheet](#)

Geometry source: hardware/mechanical/enclosure.py and reference/pcb.step. Nominal, not production-toleranced.

FIT CHECKS

Valid connected solids; closed STL meshes.
No overlaps with the board or available component references.
Some component models are missing or illustrative.
Prototype-test side snaps, solder tails and repeated lid removal.
Release: press both side windows, then lift the cover.
Snap arms: 7.5 x 0.6 x 1.2 mm; vertical play: 0.2 mm.